

Finite Network Improvements for the Exact Discrete Fourier Transform

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We give a reusable conditional transfer from finite complex tensor networks to the exact discrete Fourier transform, together with versioned network witnesses. The current proposed all-length bound is

$$T(n) = O\left(n(\log n)^{1-\delta}\right), \quad \delta = 0.00067.$$

The selected finite witness has strict tensor saving $a = 0.00067147467$. Independent mathematical review and end-to-end formal verification remain pending. This paper separates the reusable transfer theorem, the selected construction, and the history of incorporated improvements. No practical FFT speedup is claimed.

1. Scope and the finite-network interface

This paper addresses OpenAI Problem #130. Its reusable argument transfers finite tensor-network improvements to an all-length exact discrete Fourier transform. Each selected network is a separate, pinned result record. A passing numerical certificate alone is not a transfer proof. The result history records which networks have a written transfer and which remain candidates.

We use the exact-complex-arithmetic model of OpenAI’s *An explicit power saving for the exact discrete Fourier transform* [O]. Complex field operations have unit cost, coefficients and intermediate magnitudes are unrestricted, and a specified root of unity of order less than $1024n^3$ is supplied. Integer operations and random access on a fixed number of $O(\log(n+2))$ -bit words have unit cost. Scalar preparation, index construction, initialization and data movement are charged. This is not a bit-complexity, numerical-stability, bounded-coefficient or practical FFT claim.

The fixed kernel is

$$C = \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}, \quad C^2 = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Finite-network contract. Fix integers $m \geq 2$, $W \geq 1$, and nonnegative integer multiplicities n_r for $1 \leq r < m$. For every column count $f \geq 1$, require a constructively specified finite algorithm that applies $C^{\otimes mf}$ to **all W arbitrary complex input arrays**, including arrays used as dirty auxiliary registers. It must have the following properties.

1. Its positive-width operations consist of n_r calls on rf binary axes in one array, independently on every fiber of the remaining axes. Each such call is $C^{\otimes rf}$, up to explicitly implemented adapters.
2. Everything outside these calls costs at most $AW2^k$ for $k = mf + b$, $0 \leq b < m$, with arbitrary spectator axes allowed. The constant A is independent of k and f . Temporary copies, scalar gates, phase corrections, layout passes and their inverses are included.
3. The schedule and fixed tables have constructive finite preparation procedures. Serial execution uses $O(W2^k)$ live complex registers and addresses represented by a fixed number of $O(k+1)$ -bit words.
4. The specified input-output map is exact on arbitrary data and scratch. The histogram counts actual recursive calls, including copied-register transformations and endpoint corrections. A rank budget or aggregate scalar identity by itself does not satisfy this condition.

This contract makes no particular choice of number of stages, producer DAG, frame system or network optimizer. Different witnesses may discharge it by different arguments. A fixed finite cover is permitted: if a cover has V vertices and counts $W_0, n_r^{(0)}$ per vertex, the actual values are $W = VW_0$, $n_r = Vn_r^{(0)}$. The factor cancels in the normalized moment, but the algorithm still allocates and routes the full finite stock. Rational per-vertex counts require an explicitly justified common integer replication; they are not fractional physical registers.

2. A reusable transfer theorem

Conditional transfer theorem. Suppose the finite-network contract holds and, for a rational $0 < a < 1$,

$$\Psi(1-a) = \frac{1}{W} \sum_{r=1}^{m-1} n_r \left(\frac{r}{m} \right)^{1-a} < 1. \quad (2.1)$$

Then the network supplies a uniform arbitrary-input tensor algorithm of cost

$$O\left(2^k(k+1)^{1-a}\right). \quad (2.2)$$

Under [O]’s Fourier compiler and all-length reduction, every fixed $0 < \delta < a$ gives

$$T(n) = O\left(n(\log n)^{1-\delta}\right) \quad (2.3)$$

in the model of Section 1. The constants and eventual thresholds may depend on the entire finite witness. This theorem is conditional on the actual algorithmic contract, not solely on the finite inequality (2.1).

2.1 Incomplete batches and unequal child widths

Let $T(k)$ denote the simultaneous cost on W arbitrary arrays, and set $t(k) = T(k)/(W2^k)$. Handle the fixed range $k < m$ directly. For $k \geq m$, let $f = \lfloor k/m \rfloor$ and process the fewer than m leftover axes directly. A width- r call has 2^{k-rf} fibers. Pack complete groups of W fibers and recurse; apply the ordinary coordinate algorithm to the fewer than W leftover fibers. Thus

$$\begin{aligned}
T(k) &\leq AW2^k + \sum_r n_r \left\lfloor \frac{2^{k-rf}}{W} \right\rfloor T(rf) \\
&\quad + B(W-1) \sum_r n_r rf 2^{rf}.
\end{aligned} \tag{2.4}$$

Every child has $r < m$, so

$$rf 2^{rf-k} \leq rf 2^{-(m-r)f} \leq r.$$

The last line of (2.4), after normalization, is bounded by a constant independent of k . Consequently

$$t(k) \leq A' + \frac{1}{W} \sum_r n_r t(r \lfloor k/m \rfloor). \tag{2.5}$$

Put $\theta = 1 - a$. Choose K for the finite base range and so that $K(1 - \Psi(\theta))m^\theta \geq A'$. Since $rf \leq (r/m)k < k$, strong induction gives

$$t(k) \leq A' + Kk^\theta \Psi(\theta) \leq Kk^\theta.$$

A single requested transform initializes $W - 1$ additional arrays and keeps its own output. Because W is a fixed integer, this remains $O(2^k(k+1)^\theta)$. Serial children have at most half their parent's array length, so reusable buffers give a geometrically summable $O(W2^k)$ live stock. The axis count contracts by the fixed factor $\max r/m < 1$, giving $O(\log(k+2))$ stack depth.

2.2 Propagation to every Fourier length

The proof of [O], Proposition 4.2 consumes the arbitrary-input tensor bound, including scalar and address work. Substitution of (2.2) gives its working-length bound

$$O\left(R(\ell+1)^\theta(1+\log r_{\max})^4\right) + \text{poly}(\ell, r_{\max}, \log(R+2)).$$

Its exact-width Fourier compiler, denominator conditions and supplied root are unchanged. Its small-prime working lengths, CRT permutations and charged chirp convolution, with $\ell = \Theta(\log n / \log \log n)$, yield

$$T(n) = O\left(n(\log n)^{1-a}(\log \log n)^{3+a}\right). \tag{2.6}$$

The fixed convolution operand remains charged. For every positive gap $a - \delta$, eventually $(\log \log n)^{3+a} \leq (\log n)^{a-\delta}$, proving (2.3). The compiler and all-length reduction are invoked with their hypotheses, not re-proved here. Integer-multiplication bit-network constraints, stopping losses and tape precision estimates are not premises of this transfer.

3. Implementing finite adapters in the array model

The layout argument extends [O], Lemma 2.5. For fixed m , a binary basis change on each of f columns and a selection of r output slots per column can be implemented in $O(2^k)$ word and movement operations. Regard each column as a digit of fixed radix 2^m . A finite table gives that digit's contributions to source and destination addresses, including the gathered rf -bit active field. Precompute displaced tables in $O(f + 1)$ work. Traverse the digit-prefix tree while maintaining address sums; it has fewer than 2^{k+1} nodes. Each extension and leaf requires a fixed number of word operations. This includes both gathering and scattering without scanning all f columns separately for each entry.

Translations are incorporated into the tables. A fixed diagonal quadratic phase with values in $\{1, i, -1, -i\}$ on each column is handled by retaining its accumulated exponent modulo four during the same traversal. Products of these column phases and fixed scalar multipliers are therefore charged linear work. Fixed permutations of the finite role stock and finite cover vertices similarly require a fixed number of array passes. These constants may be enormous.

Finite binary linear algebra, frame representatives, gate coefficients and cover tables are computed constructively once. All of their dimensions are fixed independently of k . Neither free data movement nor free per-entry address reconstruction is assumed. A witness using other adapters must supply an additional cost argument.

4. Exact moment certificates

For a given physical histogram, write

$$\Psi(1 - a) = \sum_r \frac{rn_r}{Wm} \exp\left(a \log \frac{m}{r}\right).$$

The checker uses rational upper and lower logarithm bounds obtained by range reduction and the atanh series. For $0 \leq x < 3$,

$$\sum_{j=0}^6 \frac{x^j}{j!} \leq e^x \leq 1 + x + \frac{x^2}{2(1 - x/3)}. \quad (4.1)$$

The upper bound follows because successive terms from the quadratic term onward have ratio at most $x/3$. Lower and upper logarithm endpoints are rounded outward to a rational grid before use. Every inequality determining acceptance is rational; decimal displays are explanatory only. The earlier round-six certificate uses a looser geometric exponential bound, which remains valid and unchanged.

A result record contains its exact tensor witness a , Fourier choice δ , positive absorption gap, physical histogram, immutable source pins, verifier, and explicit proof dependencies. A lower moment above one at a larger a rejects that value for the same histogram; it is not an optimality theorem over other networks.

5. Versioning and evidence boundaries

The stable source of Sections 1–5 is `paper/framework.md`. Network-specific arguments live under `results/`; `results/registry.json` selects the current proposed result explicitly. The build assembles the selected record into this paper and generates the improvement log. It never selects the largest upstream number automatically.

An incorporated result requires a written discharge of the network contract, a pinned complete witness, exact moment and absorption checks, a reproducible finite verifier, and a review record stating its limitations. The status **proposed** does not mean independently reviewed or formally verified. Candidates remain visibly separate from incorporated results. Earlier proofs, certificates and publication artifacts remain available even after a newer result is selected.

This work is AI-assisted. Additional checks written in this repository are not independent mathematical review. Vendor checkers are identified as such; random exact or modular trials are not described as universal identities. Lean arithmetic certificates are distinguished from a formal network or Fourier theorem. No novelty or priority is claimed for imported network improvements.

The following selected witness supplies the construction-specific arguments and attribution. Its status and source pins belong to that record; future updates need not rewrite the general transfer theorem.

6. Base construction: round-ten paired cubes

Record date: October 9, 2026. **Status:** proposed conditional Fourier transfer with finite checks; independent mathematical review and end-to-end formal verification remain pending.

The complex word is pinned to Swapnil Jain’s `integer-mult-kappa`, commit `d2f6146ffb` `bd893e69b8fb06bc53b82af36d8032` [J10]. Its tensor witness and our Fourier choice are

$$a = \frac{15402419}{25000000000} = 0.00061609676, \quad \delta = \frac{61}{100000} = 0.00061. \quad (6.1)$$

This record replaces the round-six finite network. It does not add successive networks’ savings or substitute the integer-multiplication headline κ into the Fourier theorem. Its new dependencies include icekylinx’s arbitrary-subspace frames and three-stage covers [C130], paired cubes and completed-core sharing [C144], and the birth-reuse and module lineage retained in [J10]’s NOTICE.

6.1 Finite word and scalar identity

There are $p = 11$ coordinate pairs, $h = 22$ local binary coordinates and $v = 8\binom{11}{3} = 1320$ ports. A port chooses one coordinate in each of three distinct pairs; its address q_S has norm one. The eight selectors on a fixed triple of pairs form a cube. The ambient cover has $m = 3h = 66$ coordinates.

For ports T, S , define

$$B_{T,S} = \frac{|T \cap S| - 1}{2}, \quad B_{\text{block}} = \bigoplus_I B|_I, \quad K = I - B_{\text{block}}, \quad H = B_{\text{block}} - B.$$

Then $K + H + B = I$. Within each cube, $K = (P - A)/2$, where P is antipodal permutation and A is adjacency at selector distance one. Its Walsh eigenvalues are $-1, -1, 1, 1$, according to selector weight, so $K^2 = I$. The opposite-parity blocks are signed 4×4 Hadamard matrices divided by two.

The frozen signed addition DAG supplies H and the coordinate-star contribution B . It uses a searched restriction of eumemic's PR117 triple module, the unchanged G37 pair module from eumemic's PR168, nested-prefix all-but-one modules, merged face/edge reads, and compatible carrier arcs. Jain's pair-aware gauge choice and operation-frame refinement are part of the pinned word. Re-running the search is unnecessary to specify this finite witness; verifying its final word is essential.

The word has 18,259 addition nodes, 4,477 output uses, 15,132 logical auxiliary roles and 32,071 compiled operations. A coordinate-star role is read through a paid copy at frame zero while its original remains on its ascending chain. The total copied-star loss is $\ell = h(h - 2) = 440$. The source registers themselves implement K : their parity blocks meet at a common three-dimensional frame, deliver their contributions at the assigned target hyperplanes, and undo K at the full frame before source-subtraction cleanup. Each source contributes widths $[2, h - 4, 1]$; no extra K bank is free or required. These are the exact scalar and geometric interfaces of [C144], instantiated on the new word.

Our verifier expands the signed DAG as integer coefficient vectors. After multiplying output coefficients by six, it checks **every** entry of $6(H + B + K) = 6I$ and every cube's $K^2 = I$. The full $1320^2 = 1742400$ coefficient check supplements the upstream random modular decoder trials. Conservative address spans include visited inputs even if signed coefficients cancel.

6.2 Arbitrary-subspace frames and exact adapters

The round-six projector argument is insufficient for the degenerate subspaces in this word. We instead use [C130]'s preimage-frame construction. In the binary symplectic space $\mathbb{F}_2^h \oplus \mathbb{F}_2^h$, put

$$L_0 = \{(0, z)\}, \quad F = \begin{pmatrix} I & I \\ 0 & I \end{pmatrix}, \quad L_U = \{(u, u + v) : u \in U, v \in U^\perp\}.$$

Here F also denotes the exact operator $C^{\otimes h}$; its exact square is a translation, although its binary symplectic square is identity. The spaces L_U are Lagrangian, including for degenerate U , and

$$d(L_U, L_V) = \dim U + \dim V - 2 \dim(U \cap V), \quad FL_U = L_{U^\perp}, \quad (6.2)$$

where $d(L, L') = h - \dim(L \cap L')$. Indeed the intersection is parameterized by $u \in U \cap V$ and $v \in (U + V)^\perp$, and the symplectic pairings vanish.

For an invertible binary matrix G with $GE_r = U$, choose an exact Gaussian-dyadic lift of

$$K_G = \begin{pmatrix} G & 0 \\ G + G^{-T} & G^{-T} \end{pmatrix}, \quad T_U = K_G C_{E_r} K_G^{-1}.$$

The lift K_G is a binary permutation followed by a quadratic phase. In coordinates its phase matrix is $I + (GG^T)^{-1}$. Use $T_0 = I$, $T_H = F$, the inherited norm-one source representatives, and

block-factored representatives for cover offsets. The preimage convention is $L(T) = T^{-1}L_0$, so a physical transition $T_V T_U^{-1}$ has Fourier rank $d(L_U, L_V)$. For $U \subseteq V$ this is $r = \dim V - \dim U$.

The exact Clifford elimination interface of [C130] factors that transition into two rank-zero adapters and r coordinate Fourier factors. No new irrational constant is needed, since

$$\frac{1+i}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = SCS, \quad S = \text{diag}(1, i).$$

All rf coordinate factors are gathered into one recursive child. The adapters are the finite permutations and quadratic phases charged in Section 3. Registers participating in a scalar gate use the **same exact representative**; matching subspaces alone does not authorize the gate. Differently transported representatives with the same preimage are reconciled by their actual rank-zero quotient. Thus transitions telescope without assuming that distinct partial frames commute.

For the reverse/complement word use $D_U = T_U F^{-1}$. Its preimage is $L_{U^\perp}$ and $D_U D_V^{-1} = T_U T_V^{-1}$ exactly. Reverse every scalar operation, compensation read and cleanup, not merely the positive producer. Reconcile full endpoints by the actual Pauli and phase adapters. This preserves the transition widths while fixing their physical orientations.

The arbitrary-dimensional Clifford factorization is an explicit upstream proof dependency. Our exhaustive four-dimensional checks cover all 67 subspaces, the distance and complement identities, exact nested transitions, and the full-frame translation square; they do not prove the general factorization by enumeration.

6.3 Delayed compensation and physical register reuse

A reversible scalar program with source injection V , mixer M and readout J satisfies

$$JM(z + Vx) - JMz = JMVx$$

for arbitrary dirty z , and its inverse mixer and source subtraction restore z . When centre outputs are read at an intermediate clock, the same identity uses the sum of their linear responses at the actual read clocks. Back-propagating their exact coefficients through the signed gates gives the compensation coefficient for each initial role.

Reuse lemma. Suppose a recipient role has no incident operation before clock t , and its old-value read occurs at t before its first incident operation. Its initial scalar value can be any w : the output response to w is exactly its back-propagated coefficient times w , canceled by that old-value read. Now suppose a donor has finished all incident operations and is not an output role. Place the recipient on that physical register after the donor's last operation. The current donor value is a permissible w , even though it depends on earlier data and scratch. The recipient's compensation removes its entire contribution. The earlier part of the computation is unaffected because the recipient had not been touched and the donor is no longer read. This argument iterates down an acyclic chain of reuse pairs.

The actual sequence consists only of invertible gates on distinct physical registers. After the readouts, reverse that sequence at the full frame, then subtract the original source injections. This restores every initial physical scratch value. Hence reuse does not require zero scratch, an extra fresh array

or a data-dependent erasure. This is the birth/late-compensation mechanism of PR124 and PR143, specialized to the checked word.

The physical hand-off additionally requires the donor's last frame to lie inside the recipient's entrance gauge, and all later frames on the joined chain to remain nested. The donor frame moves to the recipient frame before the read. Every positive rank difference is paid. If the subspace representatives differ, their exact quotient is implemented as in Section 6.2.

For the selected witness every one of the 3,180 gauged roles is a recipient. The 15,132 logical roles therefore occupy 11,952 physical auxiliary registers. All physical chains start at a zero gauge: source-carrier loads from zero to their norm-one line are explicitly included. No unpaired gauge tail remains, and nullity-dependent sharing contributes no additional saving to this selected complex word.

The finite checker verifies matching, acyclicity through strict time ordering, final donor use, the absence of donor output roles, first recipient use, exact compensation-target support, frame containment and every joined chain. It also checks the carrier payloads at centre and final read clocks. The imported aliased scalar replay uses two exact modular trials and negative controls; the universal arbitrary-scratch conclusion uses the reuse lemma, not those trials alone.

6.4 Three-stage cover and arbitrary-array endpoints

Use [C130]'s three-stage data cover, embedded in the $m = 3h$ space of [C144]. Let $E_0 = A \perp B \perp D$ have dimensions $h, h-1, h-1$ and embed it as a nondegenerate subspace of the 66-dimensional ambient space. For each norm-one port vector q , choose the explicit orthogonal involutions $R_{12,q}, R_{23,q}$ exchanging $A \cap q^\perp$ with B or D and fixing q . Extend them by identity outside E_0 . Invocation vertices are the finite group $\mathcal{G} = O(66, 2)$; a data port (g, q) visits $gR_{12,q}, g, gR_{23,q}$.

The forward, reverse/complement, forward sequence has scalar action

$$(X, Y) \mapsto (X, Y + X) \mapsto (-Y, X + Y) \mapsto (-Y, X).$$

The transported exit preimage of each stage equals the next entrance preimage. Exact rank-zero adapters reconcile representatives, so there are no omitted positive-width interstage connectors. In E_0 the physical outputs are $(-F_0 y, F_0 T_q^{-2} x)$. Since T_q^2 is the actual binary translation by q , a paid translation, sign and bank permutation remove this source gauge. Each data bank also pays the rank-two complement call for $E_0^\perp$, obtaining the full ambient transform. No two-stage mixed-term correction is silently reused from round six.

Independently partition the ambient space into three orthogonal h -blocks P_1, P_2, P_3 . Fixed orthogonal maps H_j carry A to P_j . Route physical auxiliary register (g, r) to vertex gH_j in stage j . Each stage routing is a bijection of the full finite register stock, and stages execute serially. These are [C144]'s shared **completed** cores: each includes all compensation, copied-centre work, scalar inverse cleanup and source subtraction.

Because all physical entrance gauges are zero, a forward completed core acts as F_h on each arbitrary auxiliary stream and the reversed core as its exact inverse, with the endpoint reconciliation included. Transported into the three orthogonal blocks, these actions tensor. The inverse block differs from its forward version by a translation because F_h^2 is Pauli. The conjugated full transform and canonical ambient transform have the same symplectic action; their exact Pauli and scalar-phase difference is implemented. Thus every auxiliary input, not just the two data banks, receives the full $C^{\otimes 66f}$.

This applies the exact core-sharing and cover lemmas of [C130, C144]. The generic gauge-tail construction needs block-factored representatives; here all tails disappear only because every gauge is paired and the physical chain recount includes the replacement hand-offs. We retain compatible block-factored representatives rather than inferring physical phases from subspace dimensions alone.

6.5 Complete histogram and moment

Let H_r be the local histogram of joined auxiliary chains, including source loads and paid centre copies; let D_r and Q_r count source and target data chains. The actual calls per cover vertex are

$$n_r^{(0)} = 3(H_r + D_r + Q_r) + 2v \mathbf{1}_{r=2}, \quad W_0 = 2v + 11952 = 14592.$$

The factor three pays all three core invocations, and the final term pays both data-bank complements. There are no unpaired-gauge exterior children. The selected table is:

r	$n_r^{(0)}$	r	$n_r^{(0)}$
1	81012	11	1446
2	53466	12	2643
3	30099	13	1188
4	10314	14	2901
5	8394	15	5343
6	4788	16	1737
7	3396	18	6480
8	2742	19	1311
9	2067	20	6006
10	3396		

Unlisted widths have zero multiplicity. The recount gives

$$\sum_r r n_r^{(0)} = 961752, \quad 66W_0 - 961752 = 1320 = 2v - 3\ell, \quad \max r = 20 < 66.$$

The actual stock is $W = |\mathcal{G}|W_0$, not 14,592 arrays. The full role count has 2,159 bits. The finite orthogonal group can be constructed by enumeration of binary matrices and exact orthogonality tests; its order is

$$|O(66, 2)| = 2^{65+32^2} \prod_{i=1}^{32} (4^i - 1).$$

Its size is independent of the requested Fourier length. Fixed role permutations, scalar work and all frame adapters are charged by Section 3 with this full stock. Its constants are not claimed practical. The group factor cancels only from the normalized moment.

The local rational verifier gives

$$1 - \Psi(1 - a) > 5.5931322116649 \times 10^{-13}$$

at (6.1), using rational endpoints rather than this rounded display. At $a + 10^{-12}$ its lower moment is greater than one, rejecting that next grid point for this histogram. The positive absorption gap is

$$a - \delta = \frac{152419}{25000000000} = 0.00000609676.$$

The physical program is finite, every child is narrower, and its adapters are charged linear array passes. Sections 6.1–6.4 discharge the finite-network contract subject to the named upstream frame, cover and sharing lemmas. Sections 2–4 then give the proposed all-length saving $\delta = 0.00061$.

6.6 Reproduction, limits and attribution

Run `python3 verification/verify_round10.py` for this record, or `python3 verification/run_checks.py` for all incorporated witnesses. The new verifier checks source hashes, the exact decoder matrix, compiled carriers and compensation targets, the imported full-size schedule and histogram replay, small exhaustive exact Clifford tests, independent rational moment bounds, absorption slack and negative controls. The regenerated certificate must match `verification/certificates/round10_reference.json`.

The frozen word and checker are imported source. Our integer coefficient and frame checks are additional same-assistant implementations. We do not enumerate the full 66-dimensional cover, execute the complete DFT, rerun the network optimization search, or formally prove the infinite recurrence. The small exact frame tests do not certify every full-size physical phase. Their universal justification is the written exact-representative argument and cited factorization. The included Lean file checks numerical histogram/assembly predicates; this project’s default runner does not execute Lean, and Lean does not prove the Fourier transfer.

Independent review should prioritize the general Clifford factorization and uniform adapter cost, exact representative choices at shared gates, arbitrary-scratch birth reuse in both orientations, the shared completed-core endpoints, and the full-stock recurrence. A missing physical interface cannot be repaired by a successful moment certificate.

Jain supplies the selected word and optimization integration. The three-stage cover, paired-cube identity and frame machinery are attributed to icekylinx; completed-core sharing to an664; the DAG, modules, merged reads, physical frame descent and late-compensation lineage include eumemic; birth reuse includes jamesyc; relaxed carrier arcs include DaysSky. Douglas Colkitt maintains the community framework, with additional predecessors recorded in the preserved NOTICE. This paper supplies the proposed Fourier transfer and local checks, not authorship of those network improvements. Original licenses and AI-assistance disclosures are retained.

References for the framework and round-ten record

[O] OpenAI. *An explicit power saving for the exact discrete Fourier transform*, September 25, 2026. Lemmas 2.2 and 2.5, Proposition 2.4, Theorem 2.6, Proposition 4.2 and Section 5.4. [Public manuscript](#).

[J10] Swapnil Jain. *Integer multiplication: round ten*, commit `d2f6146ffbbd893e69b8fb06bc53b82af36d8032`. [Construction and proof boundaries](#), [frozen complex word](#), and [Lean numerical certificate](#). Research and implementation assistance is disclosed in the preserved upstream NOTICE.

[C130] icekylinx. [PR130: three-stage Cayley covers](#). The [general Clifford frames](#) and [cover proof](#) are pinned in this repository at the subsequent paired-cube commit.

[C144] icekylinx. [PR144: paired cubes and shared completed cores](#), commit `c8b22bc5c10dba497ac25804e27d9647d818e2ff`. [Paired-cube construction](#) and [exact core sharing](#). These sources record GPT-6 Astra and Codex assistance and credit an664’s PR128 sharing principle.

[BR] jamesyc’s [PR124](#) and eumemic’s [PR143](#): birth reuse and late compensation. [J10] pins and attributes their use in its construction. The module and refinement lineage also includes eumemic’s PR117, PR161 and PR168 and DaysSky’s PR162; the exact adopted data pins and notices are retained in `verification/vendor/jain_round10/NOTICE`.

7. Round-eleven refinement: retiring copies and terminal targets

Record date: October 9, 2026. **Status:** proposed conditional Fourier transfer with finite checks; independent mathematical review and end-to-end formal verification remain pending.

This record uses Swapnil Jain’s round-eleven complex word at commit `1a580dcc91dad5fbcffb5c80b7641a96bfa24e16` [J11]. It inherits the frame, paired-cube, cover and sharing interfaces of Sections 6.1–6.4, with the changed scalar word and physical ledger below. Its parameters are

$$a = \frac{67147467}{100000000000} = 0.00067147467, \quad \delta = \frac{67}{100000} = 0.00067. \quad (7.1)$$

The integer-multiplication announcement reports a different parameter, $\kappa = 0.0006685733$. We use the complex tensor witness a , together with a positive Fourier absorption gap. The bit-side cyclic all-but-one program and integer assembly are not imported into this argument.

7.1 Changes to the finite word

The dimensions remain $p = 11$, $h = 22$, $m = 66$ and $v = 1320$. The cube-local circuit changes to Jain’s C1 configuration on the attributed PR168 v4 modules. Reverse-scan carrier matching, operation-frame descent and late birth reads are retained. The new signed DAG has 21,249 addition nodes. Its 13,042 logical auxiliary roles have 2,332 gauged birth pairs, 42 deleted terminal roles and 386 hosted births. Thus

$$R_{\text{physical}} = 13042 - 2332 - 42 - 386 = 10282, \quad W_0 = 2v + R_{\text{physical}} = 12922. \quad (7.2)$$

All gauged roles are paired. Hosting introduces no unpaired gauge or exterior tail. The source-bank cube involution K , copied coordinate-star loss $\ell = 440$, source-return calls and two data-bank rank-two complements remain charged as in Section 6. The full scalar identity is checked afresh; the old coefficient and histogram certificates do not certify the changed program.

7.2 Retired-copy subtraction with arbitrary dirty contents

Suppose two retired copies P, Q carry the same input response. Move both to the **same exact** representative of a containing frame V , and execute $P \leftarrow P - Q$. The input response cancels; arbitrary dirty contents need not vanish. A fresh nongauged role B is assigned to P immediately before its first operation at V . The control Q remains a physical register and pays its climb to V

and subsequent exit. This is the zero-source-control specialization of the compatible zero-fresh reuse mechanism credited to icekylinx’s PR184 [R184]. It does not create a physically zero register for free.

The adopted hosts are disjoint triples (P, Q, B) . The birth is not a source, gauge or deleted sink; the retired copies are not output or gauged roles. Their original uses finish before the mix, the control is never used afterward, and the pivot’s subsequent uses are the renamed birth. Both prior frames lie inside V . The source record excludes pairs whose dirty part depends on a late-read value. Our complete scalar-map check below additionally verifies the actual interaction with every compensated alias; equality of abstract DAG labels alone is not the correctness argument.

Recompute the nongauged entrance compensation from the **new** transcript, including all subtraction gates. Retain each gauged role’s correctly timed suffix compensation, checking its exact target support. The initial reads are available at frame zero. The complete output-coefficient identity verifies that no residual input or dirty response is hidden in a hosted birth. Every surviving physical mixer has the form

$$z_d \leftarrow c_a z_d + c_b z_c, \quad d \neq c, \quad c_a \in \{1, -1\}.$$

After the readouts and paid moves to the full frame, reverse these actual mixers in reverse order, then subtract the original source loads. This restores arbitrary physical scratch exactly. The new inverse program, including the subtraction gates, is required.

For one host, write u_0, u_1 for the dimensions of the retired-copy frames and $s = \dim V$. In each of the three stages, the pivot loses widths $h - u_0$ and s and gains $s - u_0$; the control loses $h - u_1$ and gains $s - u_1$ and $h - s$. Zero widths are omitted. Rank mass drops by h per stage and one persistent register disappears, so the deficit is unchanged. This pays both copies’ moves. The physical verifier reconstructs the complete changed chains rather than applying this formula in isolation.

7.3 Deleting destination-only terminal registers

The 42 deletions use jamesyc’s dirty-target terminal compiler [T166], as incorporated by [J11]. A terminal role s is written only by additive updates $s \leftarrow s + b_j z_j$, is never a control, and contributes the same coefficient u to a target set T . Choose a pivot $c \in T$ with no intervening compensation read during the terminal-write interval. All writes occur after the centre-read cut. The role is neither source, gauge nor reuse participant, and the selected target sets are disjoint.

At the cut, perform $y_t \leftarrow y_t - y_c$ for $t \in T \setminus \{c\}$. Replace each terminal write by $y_c \leftarrow y_c + u b_j z_j$. After its last write, perform $y_t \leftarrow y_t + y_c$ for the same targets. If $w = \sum_j u b_j z_j$, the pivot and every other target gain exactly w , regardless of their original contents. Independent corrections to nonpivot targets survive these shears. A correction to the pivot inside the interval would be distributed incorrectly and is excluded. All target frame transitions, including these shears and replacement writes, must follow the checked descending chains.

The omitted terminal’s own initial dirty value no longer exists. Nevertheless, compensation columns for all **retained** roles must come from the full original decoder, after inserting the retirement mixes. They cannot be recomputed by deleting terminal outputs first: the replacement target writes still carry those roles’ output responses. Full source return and reverse cleanup of retained mixers remain necessary.

For a terminal at root-frame dimension r , the three-stage histogram loses 3 children at each of widths r and $h - r$, and W_0 drops by one. The full chain recount verifies these differences simultaneously for all 42 sinks, as well as pivot cleanliness and disjoint target groups.

7.4 Complete scalar map and inherited physical lift

The local checker expands every decoder coefficient over the integers and verifies $6(H + B + K) = 6I$ on all 1320^2 entries. It then separately emits the actual aliased scalar read program: entrance corrections, source loads, centre reads, delayed gauge corrections, terminal target shears, retirement subtractions, renamed births and surviving final reads.

Pulling the target coordinates backward through this program uses integer vectors scaled by six. All root coefficients have denominator dividing six; target shears are integral. The resulting map is checked to be

$$y_{\text{final}} = y_{\text{initial}} + (H + B)x$$

for **every** source coordinate, initially arbitrary physical scratch coordinate and initially arbitrary target coordinate. This is a finite coefficient identity over $\mathbb{Q}$, not a random modular trial. There are 10,282 dirty auxiliary coordinates and 1,320 coordinates in each data bank, hence 17,057,040 scalar output coefficients including zeros. Every retained mixer is checked invertible on distinct physical registers; its literal chronological inverse restores scratch after source subtraction. The separate cube check accounts for the source-bank K contribution.

For the physical lift, use the exact representatives and charged monomial adapters of Section 6.2 at every changed gate. Both retirement operands reach the same V ; each target shear and replacement write uses the event frame certified by the frozen word. The imported full-size checker reconstructs all role and target chains and width differences. The arbitrary-array lift, source-bank K execution, completed-core reversal and orthogonal routing remain the explicit dependencies from Section 6. The complete new scalar map and paid new paths are substituted into those interfaces. No literal full-size complex frame matrix or global cover is materialized by these checks.

7.5 Histogram and Fourier saving

The complete per-vertex child counts are:

r	$n_r^{(0)}$	r	$n_r^{(0)}$
1	91131	11	981
2	44583	12	3285
3	29850	13	552
4	10122	14	2229
5	7404	15	3360
6	4116	16	516
7	2706	17	69
8	3630	18	8631
9	1350	19	3201
10	2871	20	1302

They include all three completed cores and both data-bank complements. With $W_0 = 12922$,

$$\sum_r r n_r^{(0)} = 851532, \quad 66W_0 - 851532 = 1320, \quad \max r = 20 < 66. \quad (7.3)$$

The full stock is still $W = |O(66, 2)|W_0$, with the group and routing charged as in Section 6.5. At (7.1), the rational moment bound gives

$$1 - \Psi(1 - a) > 1.6106187822204 \times 10^{-12}.$$

A lower moment bound exceeds one at $a + 10^{-12}$. Exact endpoints are stored in the reference certificate; the display is rounded downward. This rejects that next grid point for this histogram, not all possible networks. The positive absorption gap is

$$a - \delta = \frac{147467}{1000000000000} = 0.00000147467. \quad (7.4)$$

The framework gives the proposed all-length saving $\delta = 0.00067$, conditional on the inherited physical interfaces and the transfer argument. This is approximately 9.18 times the previous published round-six Fourier saving and 1.10 times our incorporated round-ten choice. These ratios compare exponent parameters, not measured runtimes.

7.6 Reproduction and review boundary

Run `python3 verification/verify_round11.py`; its report must match `verification/certificates/round11_reference.json`. Source hashes, the published inventory and the Lean input histogram are checked. The upstream replay and its negative controls are identified as imported; the local checker additionally rejects a wrong subtraction sign, omitted gauge correction, missing terminal post-shear and the excessive next tensor grid point.

The default Python suite does not rerun the network search or Lean compiler. The supplied Lean file certifies numerical predicates, not this scalar compiler, the physical network or Fourier transfer. Same-assistant implementations are not independent review. General exact-frame compatibility,

cover and completed-core interfaces, the uniform cost model and the all-length reduction remain mathematical review obligations. Earlier witnesses and checks remain available independently.

Jain supplies the round-eleven integration, C1 word and selected matching. Retirement reuse is credited to icekylinx, terminal-target compilation to jamesyc, and the retained modules and frame/late-read lineage to eumemic and the other authors in [J11]’s NOTICE. Section 6 retains the original frame, cover, paired-cube and sharing credit, including an664, DaysSky, Douglas Colkitt and OpenAI. We propose the Fourier transfer and supply additional checks; we do not claim those network improvements as our own.

References added for round eleven

[J11] Swapnil Jain. *Integer multiplication: round eleven*, commit 1a580dcc91dad5fbcffb5c80b7641a96bfa24e16. [Complex construction and validation boundary](#), [frozen word](#), and [Lean numerical inputs](#). Unmodified files, licenses and notices are under [verification/vendor/jain_round11/](#).

[R184] icekylinx. [PR184: source-assisted common-frame compression](#), [proof note](#), Sections 2–3. The present word uses the zero-source-control retirement specialization described in [J11], not PR184’s numerical supplier. The note discloses substantial GPT-6 Astra and Codex assistance.

[T166] jamesyc. [PR166: dirty-target terminal compiler](#). The actual 42-sink selection and checker are pinned through [J11]. The target-shear identity is restated above; no numerical witness from PR166 is substituted into this ledger.

Incorporated result history

These are project records, not priority claims. All remain proposed transfers. Ratios compare exponent savings, not runtimes.

Date	Witness	Fourier saving
2026-10-08	legacy-shared-sum	0.00000000055
2026-10-08	round6	0.000073
2026-10-09	round10	0.00061
2026-10-09	round11	0.00067

The full log, record links and update workflow are in [IMPROVEMENTS.md](#) and [results/README.md](#). Earlier assembled papers remain under [archive/](#).